



Assignments Overdue: You can still pass! Remember, you need to pass these assignments before the course ends on July 9, 11:59 PM PDT.



THIS WEEK'S FORUM

Week 2

Discuss this week's modules here.

[Go to forum](#)

Markov chain Monte Carlo (MCMC)



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Metropolis-Hastings, Gibbs sampling, assessing convergence

Learning Objectives

Understand the basics of MCMC including Metropolis-Hastings and Gibbs sampling.

Write a statistical model in JAGS and produce posterior samples by calling JAGS from R.

▼ More

Assess MCMC output to determine if it is suitable for inference.

Module Overview



Module 2 assignments and materials 3 min

4. Metropolis-Hastings



Algorithm 9 min



Demonstration 10 min



Random walk example, Part 1 12 min



Random walk example, Part 2 16 min



Quiz:

Lesson 4 8 questions Due June 11, 11:59 PM PDT

 Code for Lesson 4

JAGS

-  Download, install, setup 3 min
-  Model writing, running, and post-processing 12 min
-  Alternative MCMC software 10 min
-  Code from JAGS introduction

5. Gibbs Sampling

-  Multiple parameter sampling and full conditional distributions 8 min
-  Conditionally conjugate prior example with Normal likelihood 10 min
-  Computing example with Normal likelihood 16 min
-  **Quiz:**
Lesson 5 8 questions Due June 11, 11:59 PM PDT
-  Code for Lesson 5 10 min

6. Assessing Convergence

-  Trace plots, autocorrelation 17 min
-  Autocorrelation 10 min
-  Multiple chains, burn-in, Gelman-Rubin diagnostic 8 min
-  **Quiz:**
Lesson 6 8 questions Due June 11, 11:59 PM PDT
-  Code for Lesson 6

 Optional Honors Content

MCMC



Quiz:

MCMC 5 questions Due June 11, 11:59 PM PDT